

Department of Electrical & Electrical Engineering
BRAC University
Semester- Spring 2026



EEE283L

Digital Logic Design

Section: 05

Design Project : Digital Elevator System

Group -2

SL	Student ID	Name
01.	24121205	Souvik Barman Ratul
02.	24121058	Muhammad Mushfiquir Rahman
03.	24121073	Abu Sadat MD. Tahsin
04.	24121236	Noureen Afroz

Submitted to

A K M Anindya Alam

Lecturer

Department of Electrical and
Electronic Engineering

Problem Statement

Modern multi-floor buildings require an efficient elevator system for transporting people between floors safely and quickly. Manual elevator control systems are inefficient because they cannot automatically process floor requests or control movement accurately. Therefore, there is a need for a digital elevator control system that can receive floor requests, process them using digital logic, and move the elevator to the selected floor automatically. In this project, a digital elevator system was designed and simulated using sequential logic circuits in Proteus. The system accepts floor requests through input switches, compares the requested floor with the current floor, controls the elevator movement direction, and stops automatically at the destination floor. The project mainly focuses on implementing digital electronics concepts such as counters, encoders, comparators, logic gates, and display systems to create a cost-efficient and reliable elevator controller.

Objective

The primary aim of this experiment is to create and implement a Digital Elevator System which utilizes various digital components and sequential circuits for performing the functions. The objectives of this experiment are as follows:

- To design an elevator system using digital logic having more than one floor.
- To learn the operation and functions of counters, comparators and sequential circuits.
- To show the requested floor number and present position of elevator on seven-segment displays.
- To control the elevator's up/down operation.
- To make a comparison between the current floor and requested floor using logic circuits.
- To simulate the whole system using Proteus software package.
- To learn practically the concept of automation.

Complex Engineering Problem Aspects

- P1 (Wide-ranging requirements): Multiple subsystems including encoding, comparison, counting, and display control.
- P2 (Depth of analysis): Requires logical design of comparator-based control and clock gating without a direct solution.
- P5 (Stakeholders): Limited to user input requirements (floor selection and correct movement).

Complex Engineering Activity Justification

- A1 (Diverse resources): Uses multiple ICs (encoder, comparator, counter, logic gates), simulation tools, and digital design knowledge.
- A2 (Interactions): Highly interactive closed-loop system where output (floor state) affects input decisions.
- A3 (Innovation): Implements automatic elevator control using pure combinational and sequential logic without microcontroller.

Requirements

Functional Requirements

The system should perform the following operations:

- Accept floor selection input from switches
- Encode the selected floor into binary form
- Compare requested floor with current floor
- Move elevator upward or downward accordingly
- Stop automatically when the destination floor is reached
- Show requested floor on display
- Show current elevator position on display
- Maintain stable operation during simulation

Non-Functional Requirements

The system should also satisfy the following conditions:

- Low hardware complexity
- Cost-efficient design
- Easy circuit implementation
- Reliable digital operation
- Accurate floor counting
- Simple user interaction

Specifications

System Specification

This digital elevator control system automates movement using sequential logic circuits all simulated within a +5V Proteus environment. By utilizing a combination of encoders, counters, and comparators, the system evaluates user requests against the current floor to determine the necessary direction of travel. Integrated seven-segment displays provide immediate visual feedback of both the destination and the elevator's real-time position, resulting in a cost-effective and practical demonstration of digital automation and electronic logic.

- Supports up to 9 floor requests
- Uses sequential logic for elevator control
- Operates with +5V DC supply
- Simulated using Proteus Design Suite
- Displays requested and current floor positions
- Uses digital logic ICs instead of microcontrollers
- Provides automatic stop at destination floor
- Supports upward and downward movement control

Component Level Specifications

Equipments

- Priority Encoder IC (74147)
- Up Down Counter IC (74LS190)
- Decoder IC (7447)
- Seven Segment Display
- Logic Gates (AND, OR, NOT)
- Logicstate
- Clock Pulse Generator
- Connecting Wires
- Digital Logic Components and ICs
- Computer or Laptop
- Proteus Simulation Software

1. Priority Encoder(74147)

Here we make a design to convert decimal floor inputs into BCD signals. It accepts multiple floor requests and generates encoded binary output for further processing in the control circuit.

2.74LS190 Up/Down Counter

The 74LS190 synchronous counter is used to store and update the current elevator floor position. It can count upward or downward depending on the movement direction signal generated by the control logic.

3.74LS85 Comparator

The comparator compares the requested floor value with the current elevator position. It determines whether the elevator should move upward, downward or stop at the destination floor.

4.74LS47 BCD to Seven Segment Decoder

The 74LS47 decoder converts BCD signals into seven-segment display outputs. It is used to display both the requested floor and the current elevator floor number.

5.Logic Gates (AND, OR, NOT)

Logic gates are used to create movement control conditions and stopping logic. These gates help generate proper control signals for sequential operation.

6. 7 Segment Displays

The seven-segment displays provide visual indication of the requested floor and current elevator position during operation and testing.

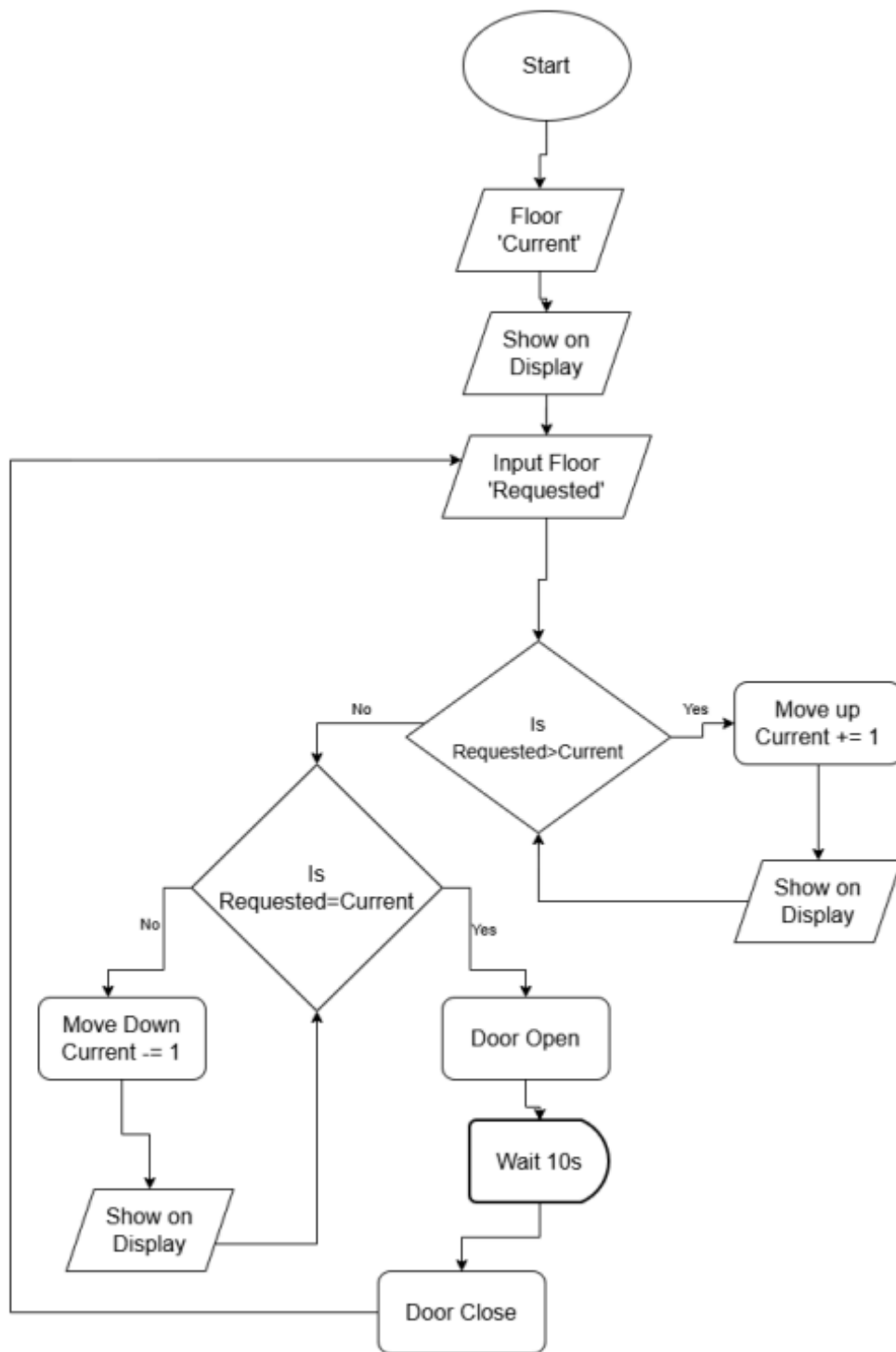
7.Clock Pulse Generator

The clock generator provides timing pulses for sequential counting operation. The counter updates its state according to the incoming clock signal.

8.Input Logic Switches

Logic state switches are used as floor request buttons. Users can select any desired floor through these digital inputs.

FlowChart

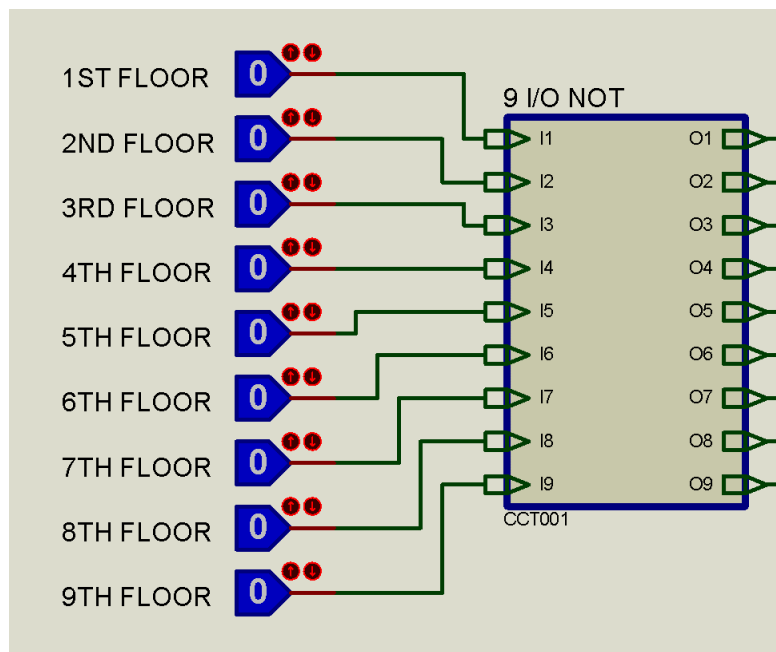


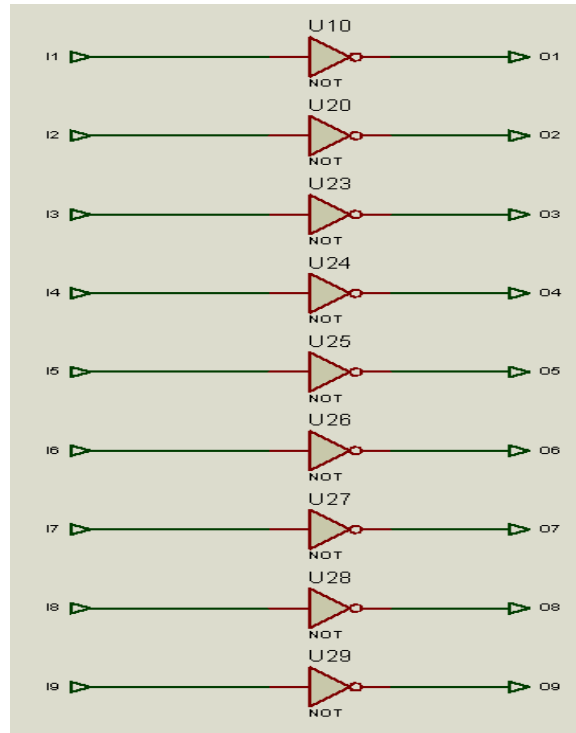
Methodology

The Digital Elevator System was designed and simulated by following several important steps. Each part of the system was built carefully to ensure proper elevator operation. The methodology of the project is explained below:

Input Section

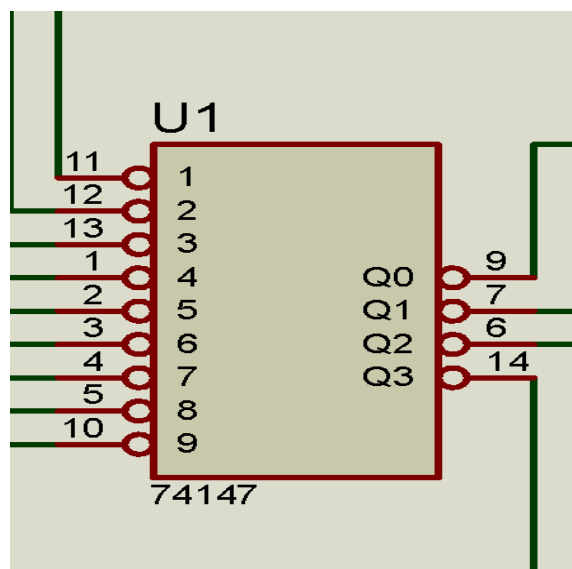
The input section was created using switches. Each switch represented a different floor of the building. When a user selected a floor, the switch generated a digital signal. This signal worked as the floor request for the elevator system. This section is important because it allows users to communicate with the elevator and select their desired destination floor.





Priority Encoder Section

An encoder IC was used to convert the selected floor number into binary form. Since digital systems operate using binary data the encoder helped simplify the processing of floor information.

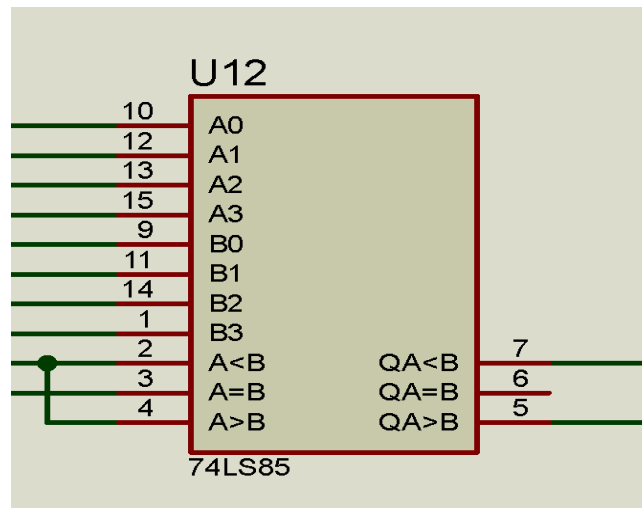


Comparator and Control Logic Section

The control section was designed using logic gates and counters. This section compared the current floor with the requested floor and decided the movement direction of the elevator.

- If, Requested Floor > Current Floor → Upward movement signal activated
- If, Requested Floor < Current Floor → Downward movement signal activated
- If, Requested Floor = Current Floor → Stop condition activated

This section acted as the brain of the elevator system because it controlled all movement operations.



Sequential Counter and Floor Tracking Section

A counter IC was used to increase or decrease the floor value step by step. The counter helped the elevator move realistically from one floor to another instead of jumping directly to the destination.

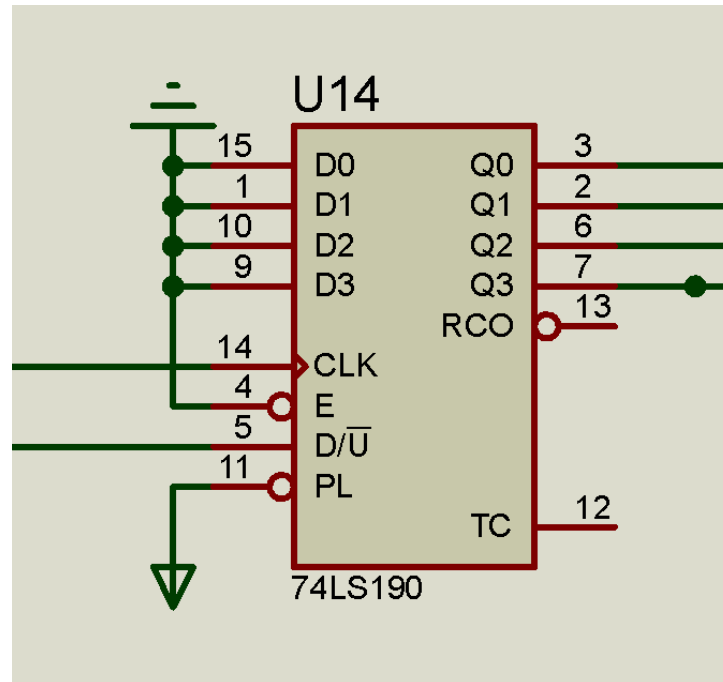
For upward movement, the counter increased its value:

0 → 1 → 2 → 3

For downward movement, the counter decreased its value:

5 → 4 → 3 → 2

This made the system behave like a real elevator.



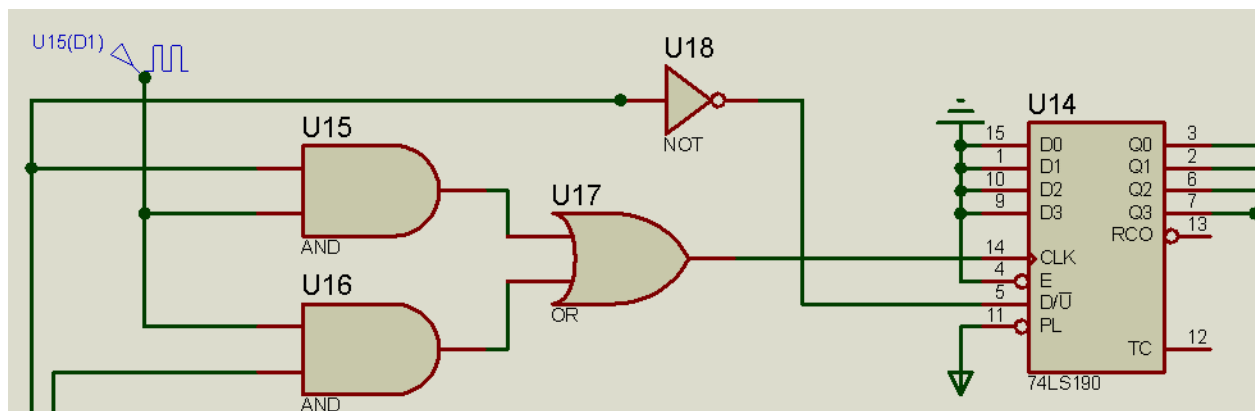
Clock Pulse Section

A clock generator was connected to the counter circuit. The clock pulse controlled the speed of elevator movement.

For every clock pulse:

- The elevator moved one floor.
- The display updated automatically.

Without the clock pulse, the elevator would not move step by step properly.



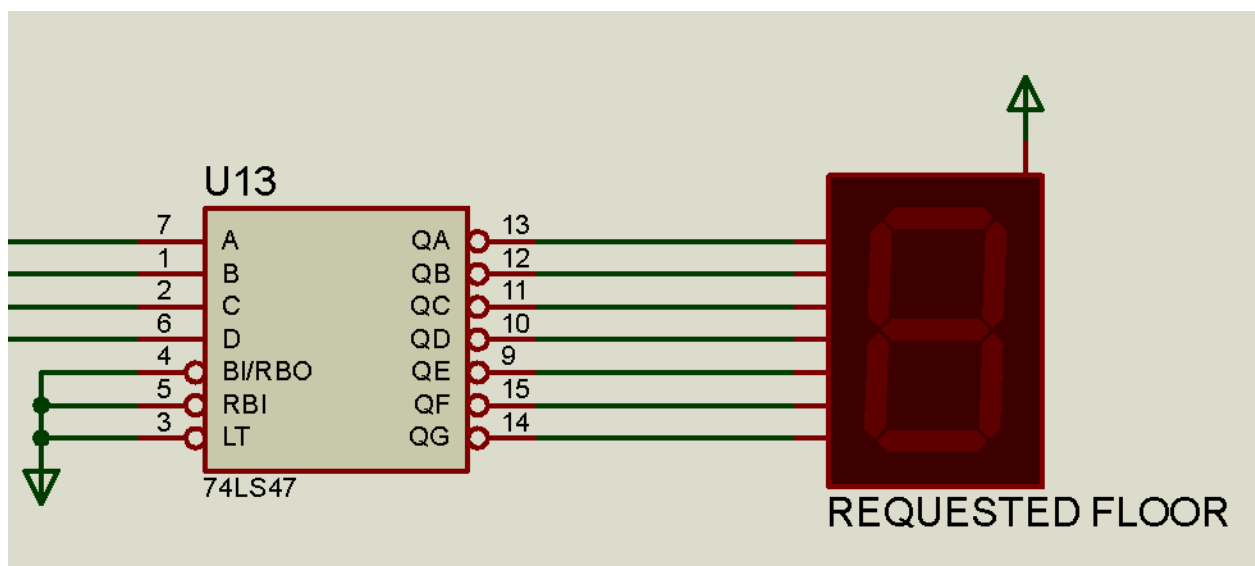
Display Section

Two seven-segment displays were used in the project. Decoder ICs(7447) convert binary values into display signals.

The displays showed:

- Requested floor number
- Current elevator floor number

This section helped users monitor the operation of the elevator clearly during simulation.



Simulation and Testing

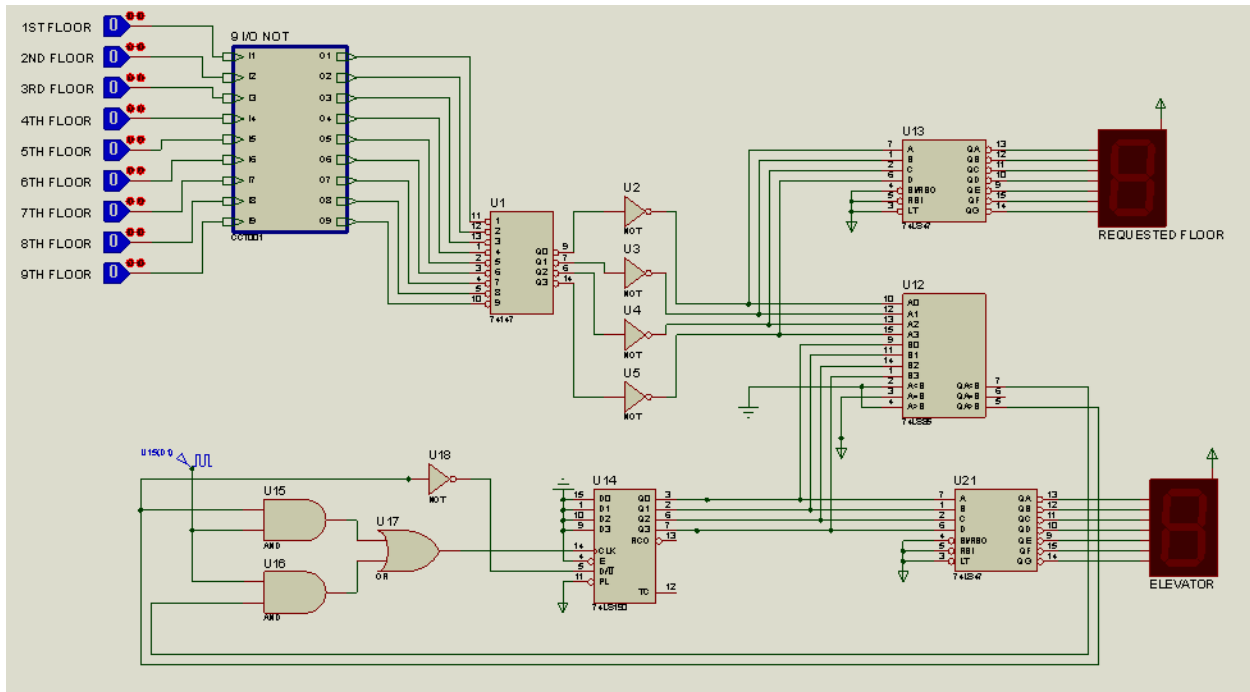
After completing the circuit design, the whole system was simulated in Proteus software. Different test cases were performed to check the accuracy and stability of the system.

The following conditions were tested:

- Upward movement
- Downward movement
- Correct floor display
- Automatic stopping condition

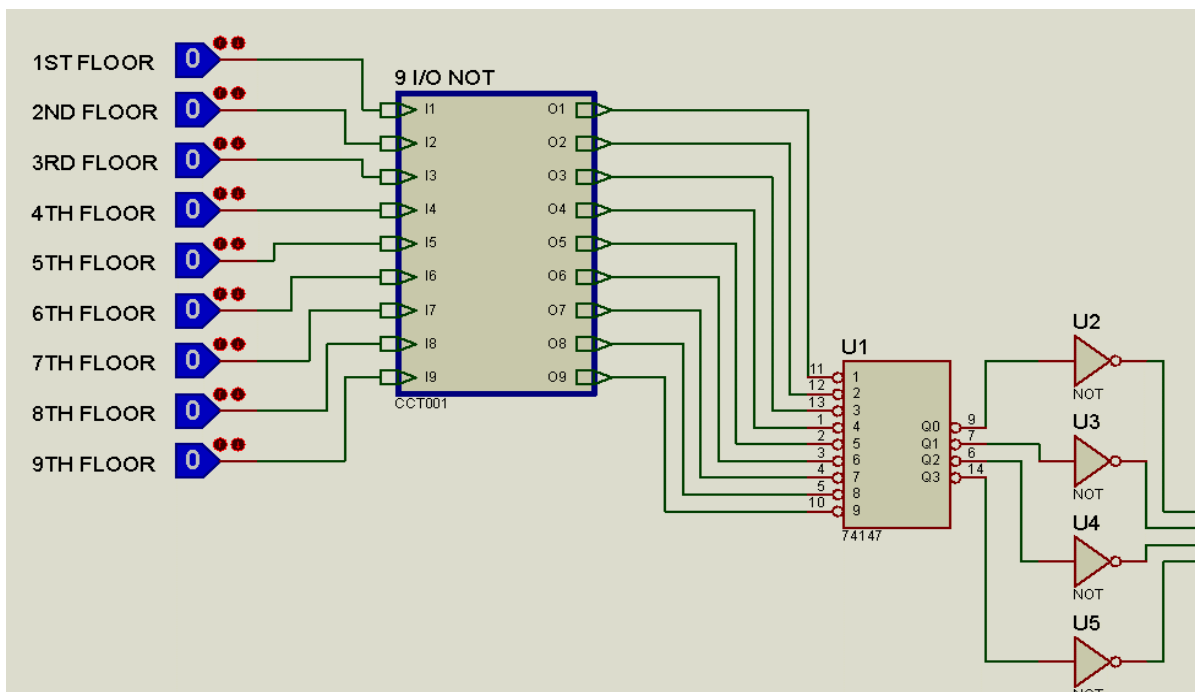
The outputs were observed carefully, and the system operated successfully according to the project requirements.

Full Circuit Diagram

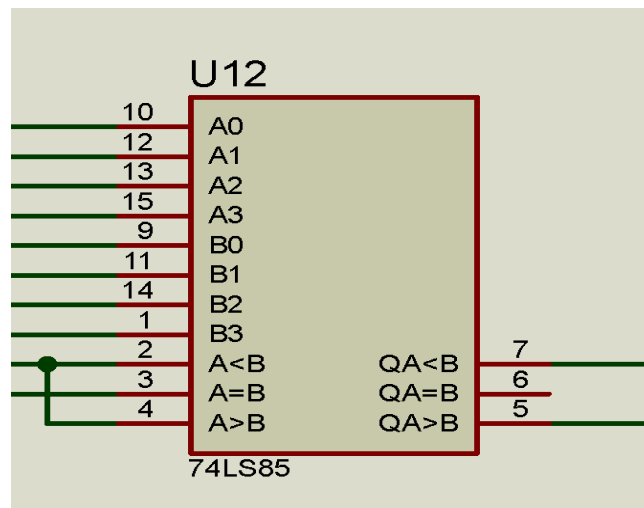


The Full System is Divided into 4 Subsystems

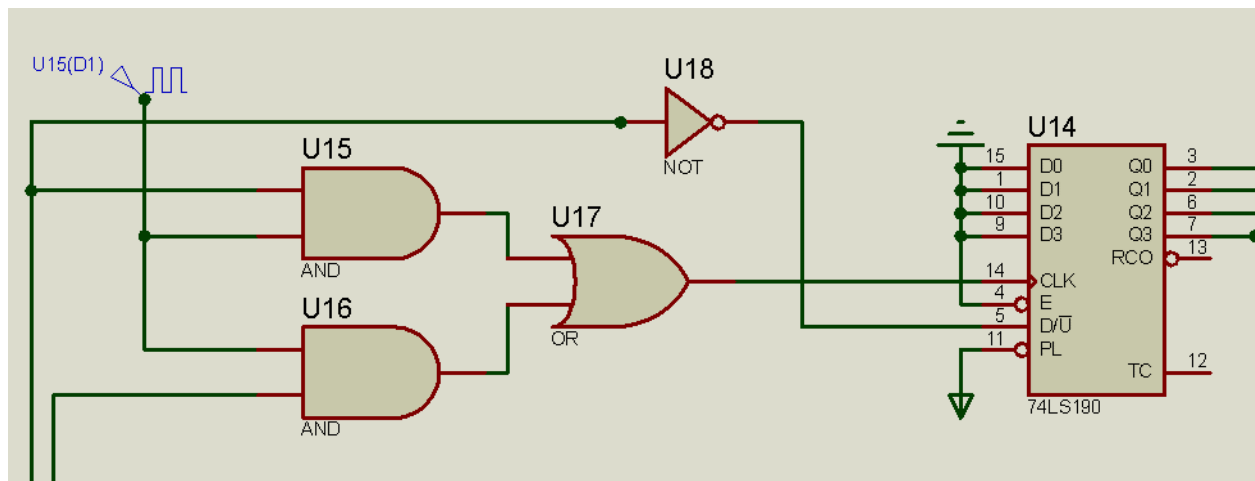
1. Input & Encoding Subsystem (Floor Selection)



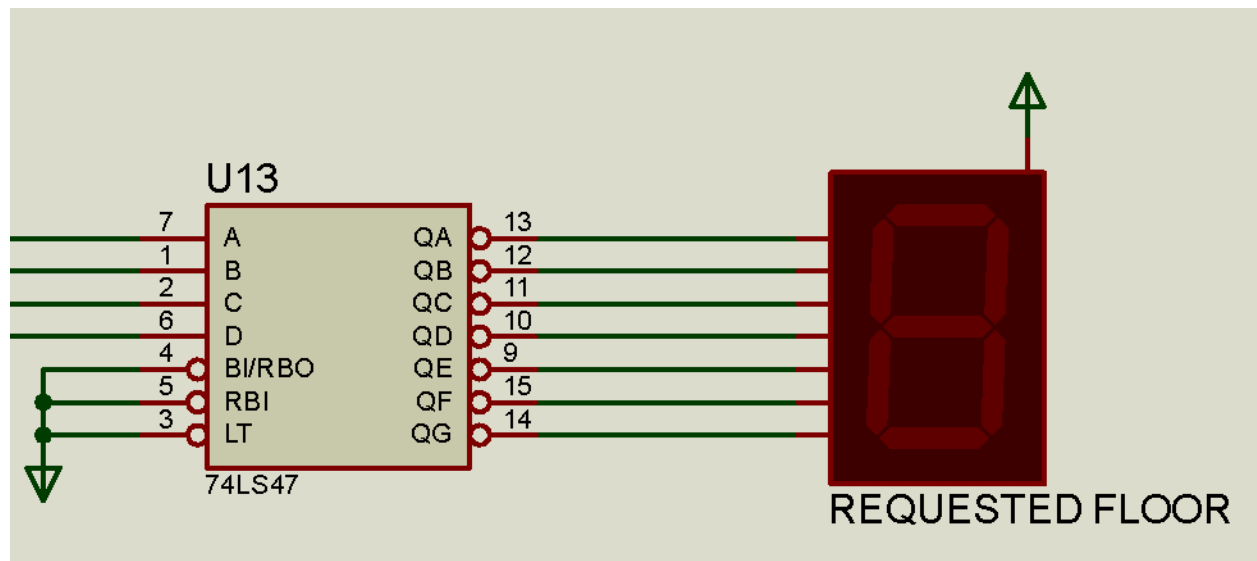
2. Comparison Subsystem (Decision Making)



3. Control Subsystem (Movement Logic)



4. Display Subsystem (Visual Feedback)



Explanation of the Solution

(1)

The elevator system uses 9 separate input buttons representing floors 1 through 9. Each button is responsible for generating a floor request signal when pressed by the user. The system also considers the condition where all inputs remain LOW as the ground floor (floor 0). These input signals are connected to a 9-to-4 encoder, which converts the selected floor request into a 4-bit binary output for further processing in the system.

Before entering the encoder, all input lines pass through NOT gates. This is necessary because the encoder used in the circuit operates with active-low inputs. An active-low encoder recognizes a logic LOW signal as an active or selected input rather than a logic HIGH signal. Therefore, NOT gates are used to invert the button outputs. When a button is pressed, the NOT gate converts the HIGH signal into a LOW signal, allowing the encoder to identify the selected floor input correctly.

(2)

In the control block, the circuit will compare the desired floor and the present location of the elevator.

Moving Upwards

When a user selects a floor higher than the current floor, the encoder converts the selected floor into a 4-bit binary value and sends it to the comparator as input A. The current floor value from the 74LS190 decade counter is connected to the comparator as input B. Since the requested floor is higher, the comparator activates the $A > B$ output. This $A > B$ signal is connected to an AND gate along with the clock pulse signal. Because both inputs of the AND gate become HIGH, the clock pulses are allowed to pass through the gate. The output of this AND gate then goes through an OR gate and reaches the clock input of the 74LS190 counter. As a result, the counter starts changing its value with each clock pulse.

At the same time, the $A > B$ output is connected to a NOT gate before entering the D/U' pin of the counter. Since $A > B$ is HIGH, the NOT gate produces a LOW signal

at the D/U' pin, configuring the 74LS190 for upward counting mode. Therefore, the counter increases its binary output step by step, representing the elevator moving upward floor by floor. The updated counter output is continuously fed back to the comparator as the current floor value. Once the current floor becomes equal to the requested floor, the comparator outputs $A > B$ and $B > A$ both become LOW, stopping the clock pulses and automatically stopping the elevator.

Moving Downwards

When the selected floor is lower than the current floor, the comparator detects that input B is greater than input A and activates the $B > A$ output. This $B > A$ signal is connected to another AND gate together with the clock pulse signal. Since both inputs of this AND gate become HIGH, clock pulses pass through the gate. The output then travels through the OR gate and reaches the clock input of the 74LS190 counter, allowing the counter to operate.

During this condition, the $A > B$ output remains LOW. After passing through the NOT gate, the D/U' pin receives a HIGH signal, which configures the 74LS190 for downward counting mode. As clock pulses continue entering the counter, the binary output decreases one count at a time, representing the elevator moving downward floor by floor. The counter output continuously updates the comparator with the current floor value. When the current floor finally matches the requested floor, both $A > B$ and $B > A$ outputs become LOW, blocking the clock pulses from the AND gates. As no clock signal reaches the counter anymore, the elevator stops automatically at the desired floor.

(3)

Floor Display

Two seven-segment displays were employed in this project:

- Requested Floor Number Display
- Current Elevator Floor Number Display

Binary numbers were decoded to display signals by the decoder(7447) integrated circuit (IC).

Test Cases1

From ground to 5th Floor

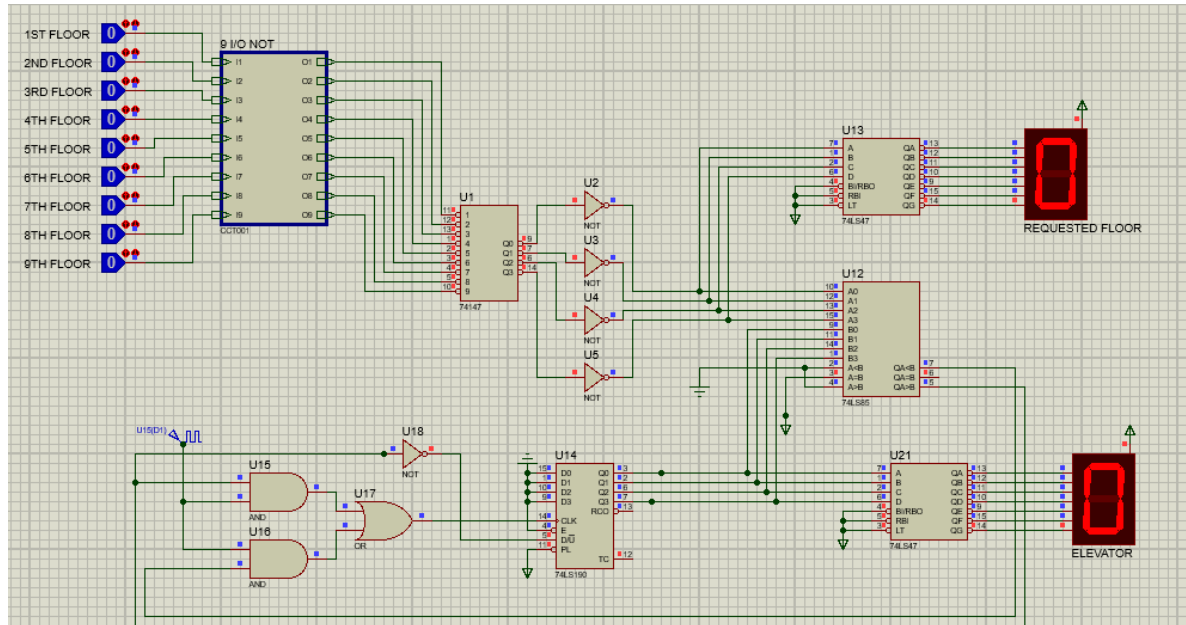


Figure: Initial Stage

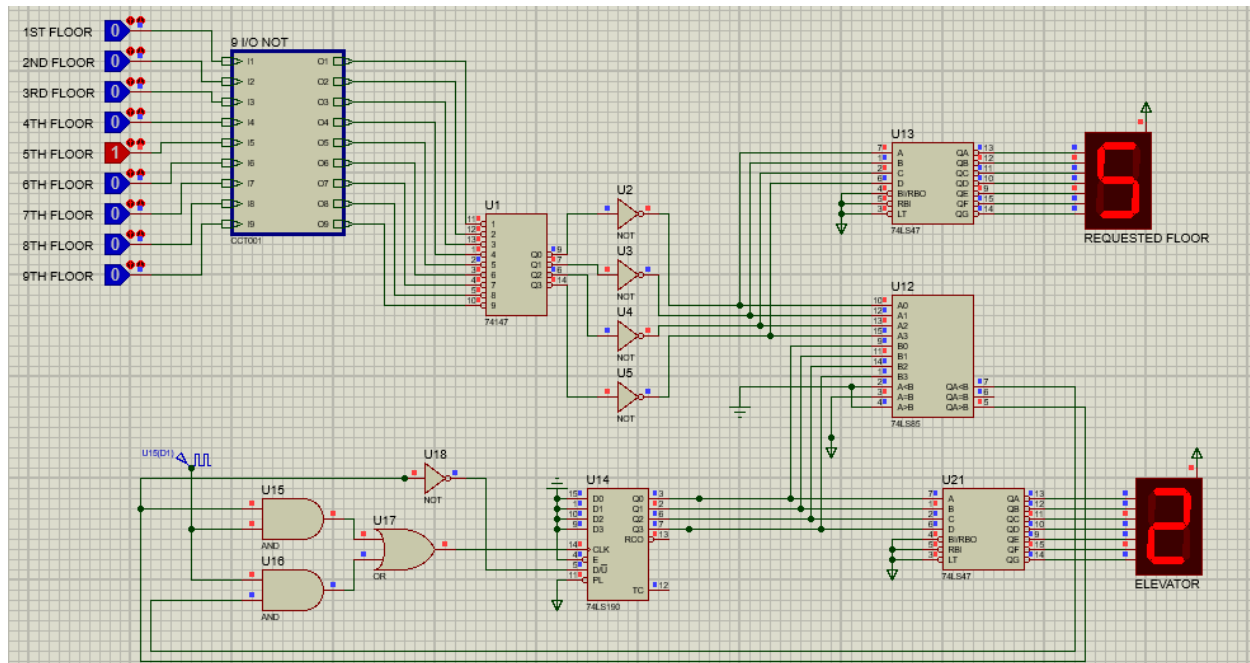


Figure: Lift moving Upward [C.F= 2, R.F=5]

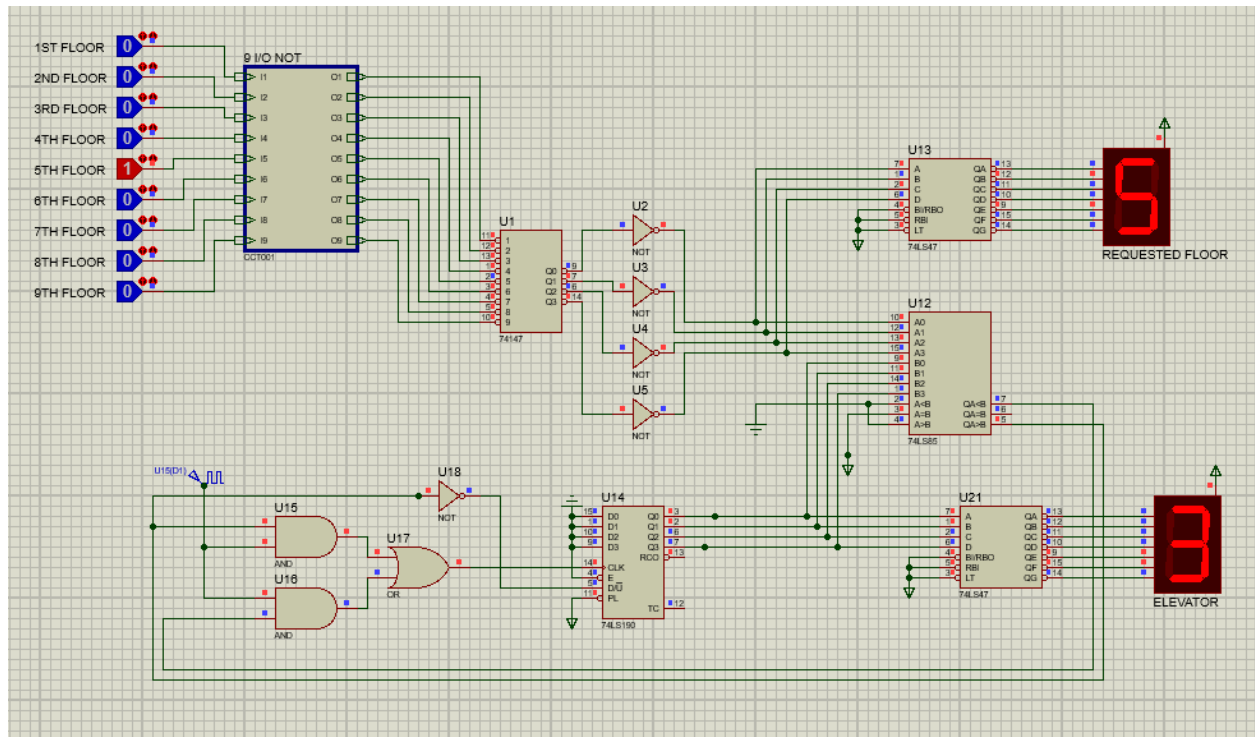


Figure:Lift moving Upward [C.F= 3, R.F=5]

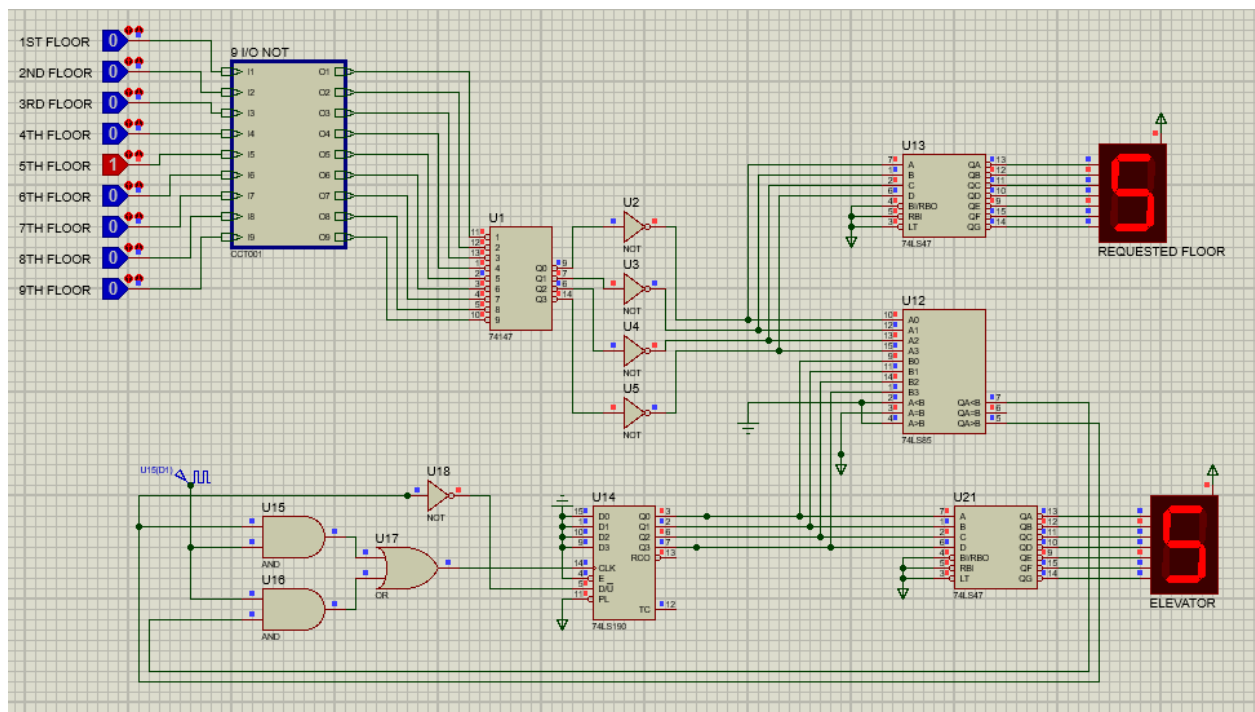


Figure:Lift reached in his destination floor [C.F= 5, R.F=5]

Test Cases2

From 7th to Ground Floor

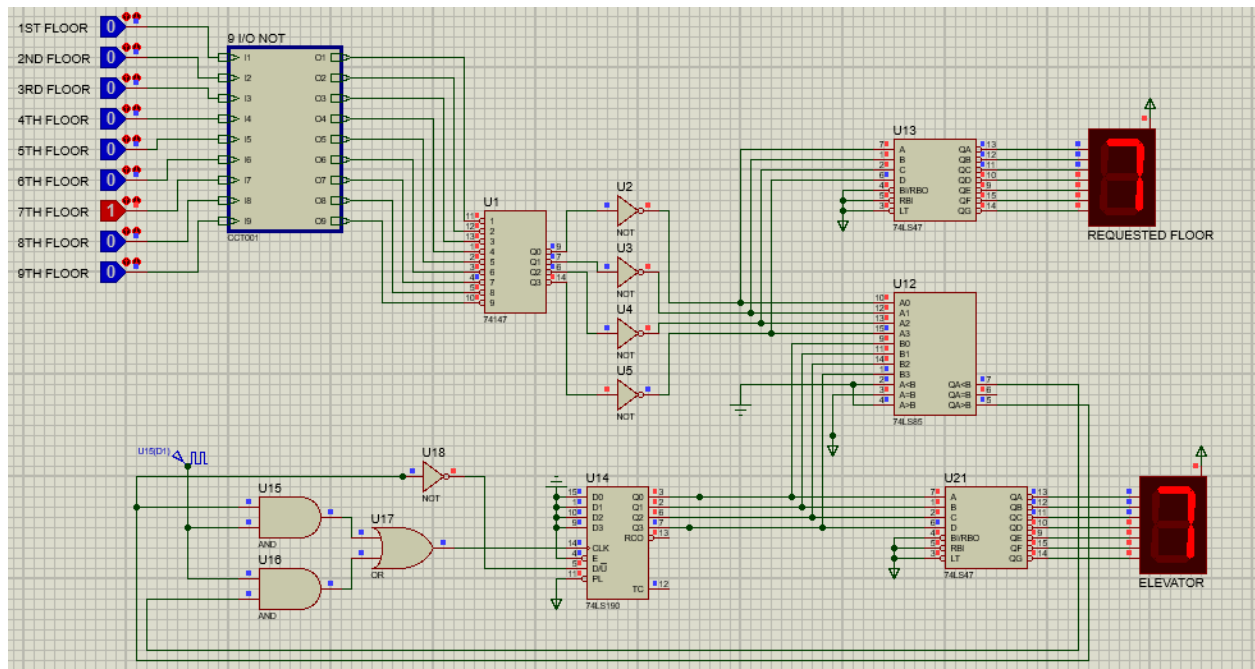


Figure: Lift Initial stage[C.F= 7, R.F=7]

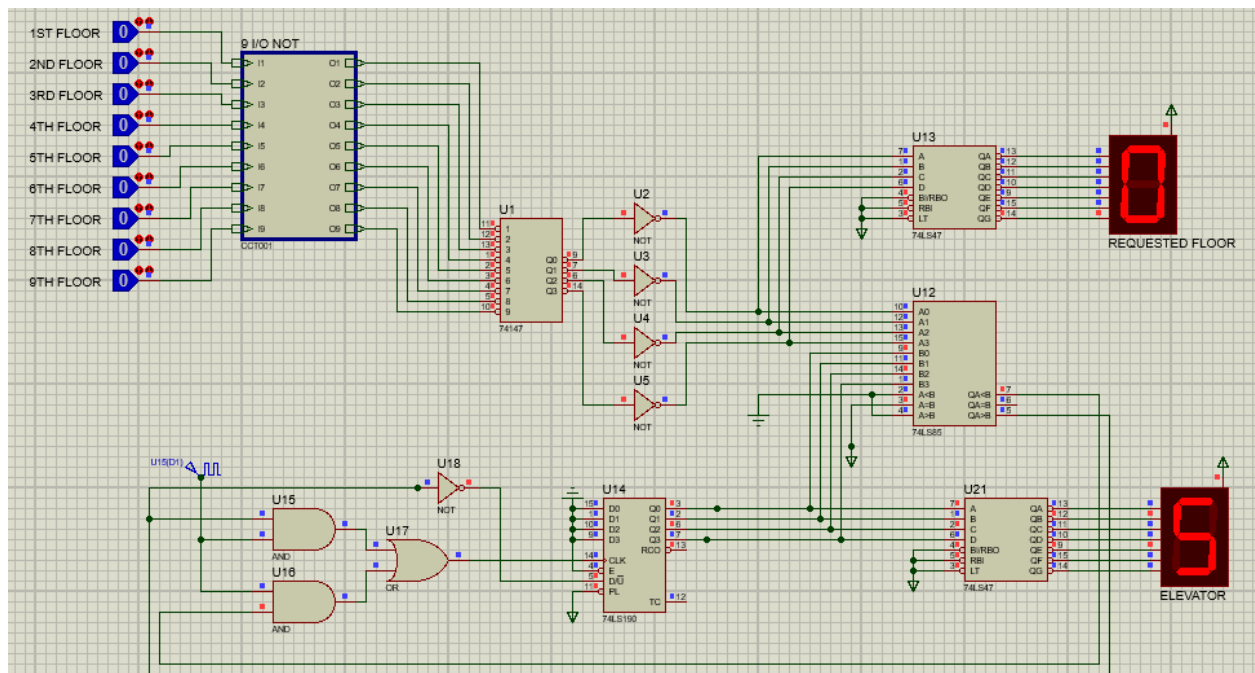


Figure: Lift moving downward [C.F= 5, R.F=0]

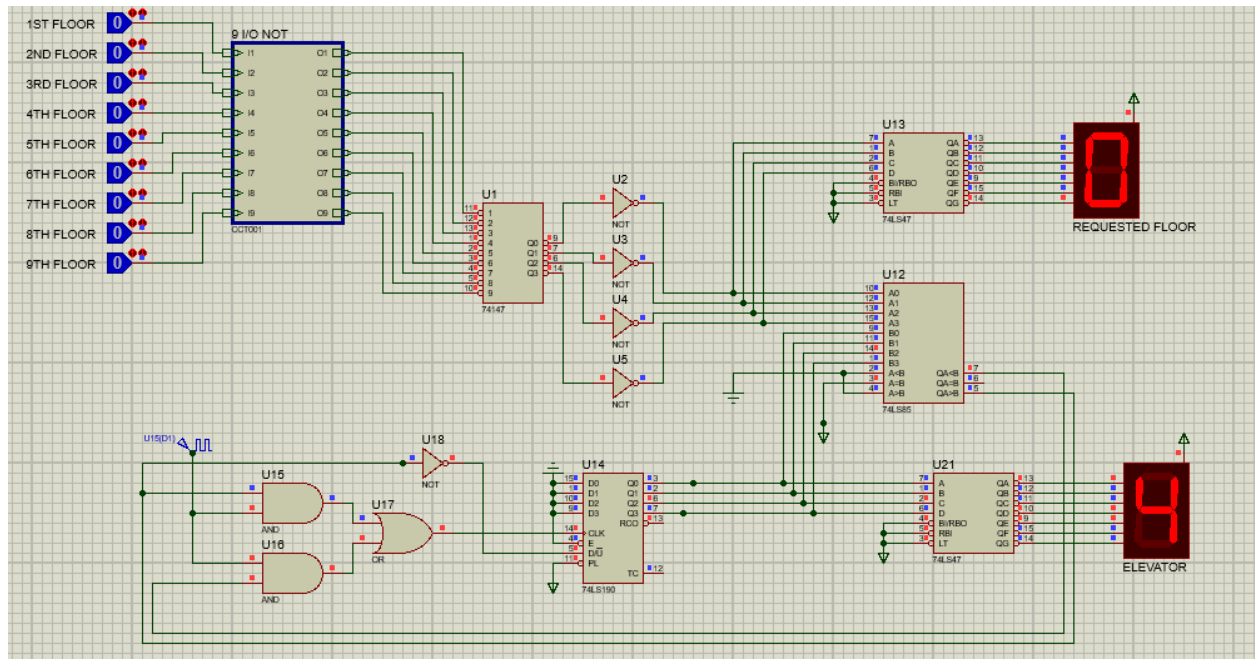


Figure:Lift moving downward [C.F= 4, R.F=0]

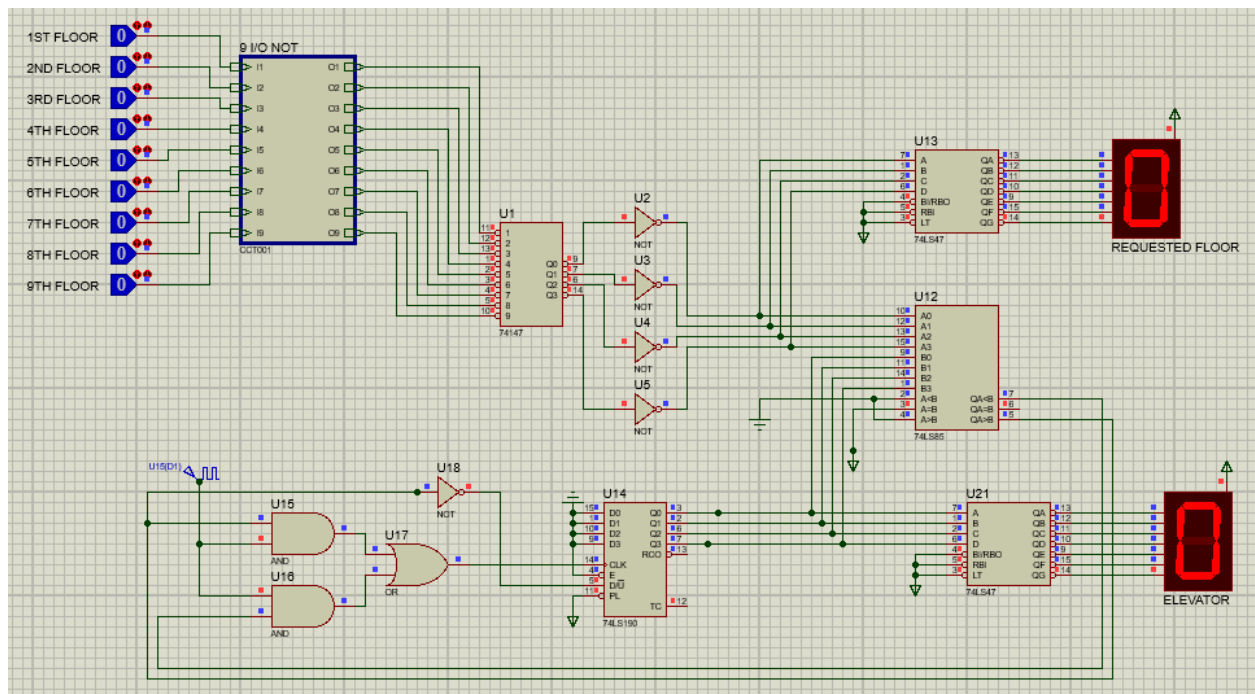


Figure:Lift moving downward [C.F= 0, R.F=0]

Evaluation of results

The Digital Elevator System is tested for various floor requests, and it is found that the system is working perfectly according to its design. The elevator successfully traverses the floors, displays correct floor numbers, and stops at the selected destination floor automatically. The switches used as inputs work perfectly, and the encoder successfully converts the selected floor number to its binary equivalent. During the upward motion of the elevator, the counter increases automatically after each clock pulse, and moves through the floors perfectly. Likewise, during the downward motion of the elevator, the counter decreases appropriately, and the elevator reaches the selected destination floor perfectly. The seven-segment displays work successfully and display the required data, i.e., both the selected floor number and the current floor of the elevator.

There were no errors in logic gate operations, and the elevator did not skip floors during its operation. The termination criterion was also working properly because the elevator terminated on completion of the specified floor. Although the project was successful, some delays occurred due to clock pulses in the simulation process. Furthermore, there were no additional features incorporated in this project, including alarm system, safety protection, door control, and sensor systems.

In conclusion, the current project was a success, as it fulfilled all the objectives stated at the beginning. It enabled us to understand the application of digital logic circuits, counters, encoder, decoder, and sequential logic in real life.

Conclusion:

The Digital Elevator System was successfully designed and simulated using sequential logic circuits in Proteus, meeting all its primary objectives. By integrating components like priority encoders, counters, and comparators, the system demonstrated a cost-efficient and reliable method for automating multi-floor transport without the need for a microcontroller. Testing confirmed that the elevator accurately processes floor requests, updates real-time positions on seven-segment displays, and automatically stops at the correct destination. This project effectively illustrates the practical application of digital logic concepts in creating functional, automated engineering solutions.

